APPM 2360 Project 1 Spring 2025

Ryan Koh, Bryce Kim, Dante Fan
University of Colorado Boulder

Contents

1 Introduction

We are modeling the population dynamics of the fictional species tribbles from the Star Trek series in a closed environment. Tribbles reproduce quickly but are hunted by a different local species called Klingons, modeling natural growth and external hunting pressure. In order to analyze the long term behavior of different tribble populations and subspecies under these conditions, we use a logistic nonlinear differential equation dependent on the tribble population size. Our goals are to simulate and understand how different tribble subspecies react to hunting, identify equilibrium solutions, and identify the ideal subspecies and starting population of the tribbles. We will model these systems, direction fields, and plots using MATLAB.

In this project, we model the population dynamics of tribbles, a fictional species, in a closed environment near the Klingon homeworld. Tribbles reproduce rapidly but are hunted by Klingons, introducing both natural growth and external harvesting pressures. To explore the long-term viability of tribble populations under these conditions, we use first-order nonlinear differential equations, specifically a modified logistic model with a harvesting term dependent on population size.

Our goals are to:

Understand how different tribble species (with varying crowding coefficients) respond to harvesting,

Identify equilibrium solutions and their stability,

Simulate population trajectories under constant and seasonal harvesting,

Recommend optimal strategies for maintaining a healthy tribble population.

We use MATLAB to simulate the system using ode45, generate direction fields, and visualize solution behavior. This allows us to evaluate outcomes under various initial conditions and harvesting intensities, ultimately guiding our decision on which species to stock and how many tribbles are needed to ensure long-term survival.

2 Analysis

See Appendix A for Lengthy Work and Code

1. Classifying and Interpreting Terms

Given the equations:

$$\frac{dy}{dt} = f(y) = ay - by^2 - H(y), \quad \text{where hunting term:} \quad H(y) = \frac{py^3}{y^3 + q}$$

Classification:

- First-order (since only the first derivative of y is in this equation)
- Nonlinear (due to y^2 and $H(y)$ in this equation)
- Autonomous (right-hand side depends only on y, not on t)

Interpretation of Terms:

- ay (Natural exponential growth of the tribble population)
- $-by^2$ (Logistic limiting factor due to resource constraints)
- $-H(y)$ (Hunting from Klingons, which depends on y)

2. Interpretation of $H(y)$

$$H(y) = \frac{py^3}{y^3 + q}$$

As $y \to 0$:

$$H(y) \to 0$$

Interpretation: When the tribble population is small, Klingons hunt few or no tribbles—since there is less of them, it is harder to hunt them due to scarcity.

As $y \to \infty$:

$$H(y) \to p = 1.5$$

Interpretation: The hunting rate hits a maximum value p, showing a limit to how much Klingons can hunt regardless of the tribble population.

Conclusion: The function $H(y)$ makes biological sense—it starts near zero and approaches a maximum value as y increases.

3. Units of Parameters a and b

- y is in *hundreds of tribbles*
- t is in *days*
- Thus, $\frac{dy}{dt}$ has units: *hundreds of tribbles per day*

From:

$$\frac{dy}{dt} = ay - by^2$$

Units:

$$[a] = \frac{1}{\text{day}}$$
$$[b] = \frac{1}{(\text{hundreds of tribbles}) \cdot \text{day}}$$

4. Analytical Solution and General Separability

With $p = 0$ the ODE reduces to a logistic equation:

$$\frac{dy}{dt} = a\,y - b\,y^2 = a\,y\left(1 - \tfrac{b}{a}y\right).$$

Separate variables:

$$\int \frac{dy}{y\,(1 - \frac{b}{a}y)} = \int a\,dt.$$

By partial fractions,

$$\frac{1}{y(1-ky)} = \frac{1}{y} + \frac{k}{1-ky}, \quad k = \frac{b}{a},$$

so

$$\int \left(\frac{1}{y} + \frac{k}{1-ky}\right) dy = a\,t + C.$$

Exponentiating and solving for $y(t)$ gives the standard form:

$$y(t) = \frac{\frac{a}{b}}{1 + \left(\frac{a}{b\,y_0} - 1\right) e^{-a\,t}}.$$

For $H(y) \neq 0$, the ODE

$$\frac{dy}{dt} = a\,y - b\,y^2 - \frac{p\,y^3}{y^3 + q}$$

remains separable:

$$\int \frac{dy}{a\,y - b\,y^2 - \frac{p\,y^3}{y^3+q}} = t + C,$$

but generally does not reduce to elementary functions.

5. Equilibrium Solutions of the ODE

An equilibrium y^* satisfies

$$f(y^*) = a\,y^* - b\,(y^*)^2 - \frac{p\,(y^*)^3}{(y^*)^3 + q} = 0.$$

Thus the constant solutions are the roots of

$$a\,y - b\,y^2 - \frac{p\,y^3}{y^3 + q} = 0.$$

These can be found numerically for each $b \in \{0.005,\ 0.05,\ 0.10\}$.

6. Graphical Analysis of Equilibrium Solutions

To analyze the equilibrium solutions of the differential equation

$$\frac{dy}{dt} = f(y) = ay - by^2 - \frac{py^3}{y^3 + q},$$

we plot $f(y)$ as a function of y for three different values of b:

$$b \in \{0.005,\ 0.05,\ 0.10\}, \quad \text{with fixed } a = 0.75,\ p = 1.5,\ q = 1.25.$$

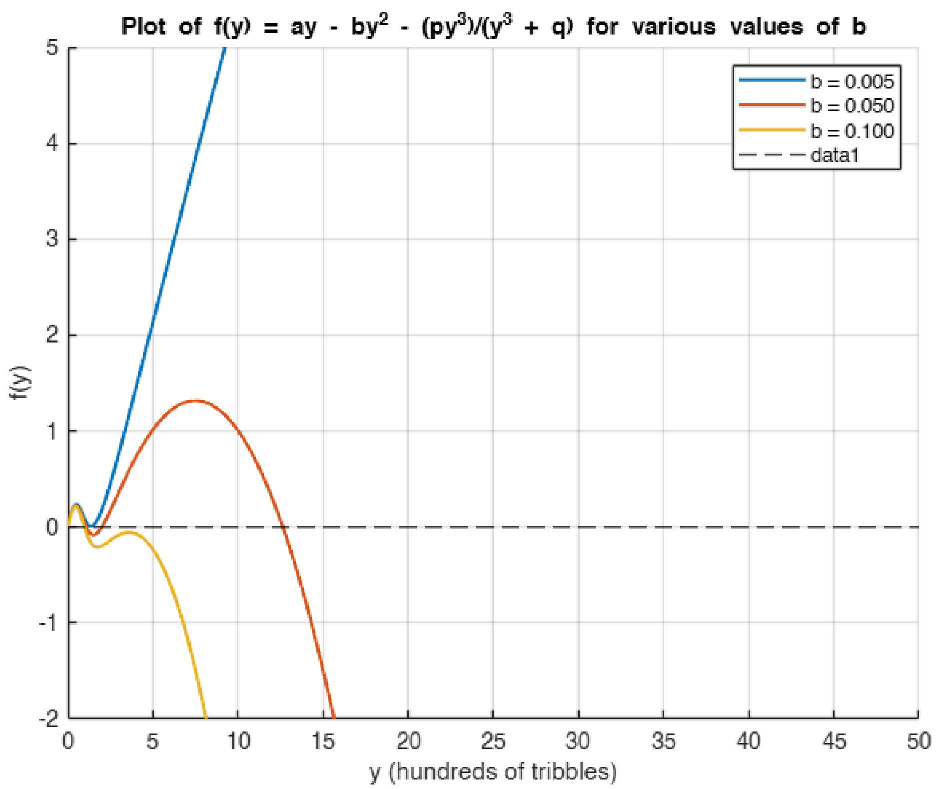

Figure 1: Plot of $f(y)$ for different values of b. Equilibrium points occur where $f(y) = 0$.

Equilibrium points correspond to the roots of $f(y)$, that is, where the curve intersects the horizontal axis ($f(y) = 0$). We make the following observations:

- **For** $b = 0.005$: There are three equilibrium points — one at $y = 0$ and two positive roots. This indicates the presence of both stable and unstable positive population levels.
- **For** $b = 0.05$: There are two equilibrium points — at $y = 0$ and one positive root. The increased density-dependent limitation reduces the number of viable equilibria.
- **For** $b = 0.10$: Only one equilibrium exists at $y = 0$, indicating eventual extinction under high population constraint.

As b increases (i.e., the species' reproductive effectiveness decreases), the number of positive equilibrium points declines. This highlights how species with low b values (high reproductive potential) are more likely to maintain a stable population despite hunting, while species with high b may go extinct due to the combined effects of density limits and predation.

Question 7

We consider the autonomous ODE

$$\frac{dy}{dt} = a\,y - b\,y^2 - \frac{p\,y^3}{y^3 + q},$$

with parameters $a = 0.75$, $p = 1.5$, $q = 1.25$, and three values of b from Question 6: $b = 0.005$, 0.05, and 0.10. For each b we

1. plotted the direction field on the rectangle $[0, 100] \times [0, 200]$;
2. located the equilibrium solutions by solving $a\,y - b\,y^2 - \frac{p\,y^3}{y^3+q} = 0$;
3. chose three initial conditions (one below the smallest equilibrium, one near the middle equilibrium, and one above the largest equilibrium);
4. overlaid the corresponding solution curves obtained via `ode45`.

Plots

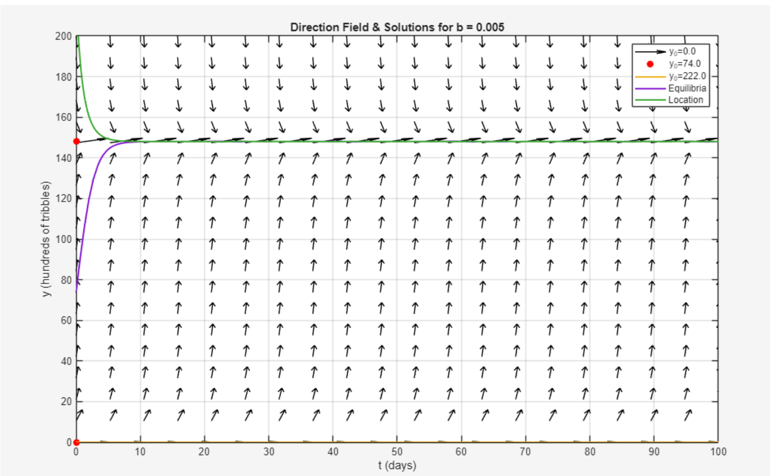

Figure 2: Direction field and solution trajectories for $b = 0.005$.

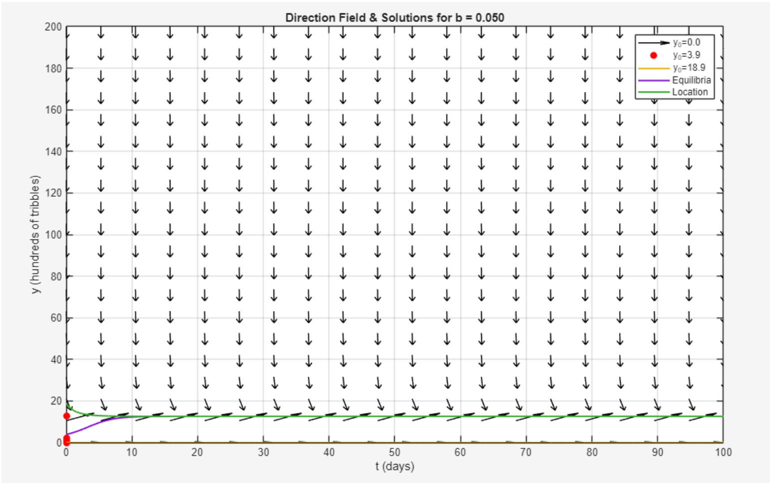

Figure 3: Direction field and solution trajectories for $b = 0.05$.

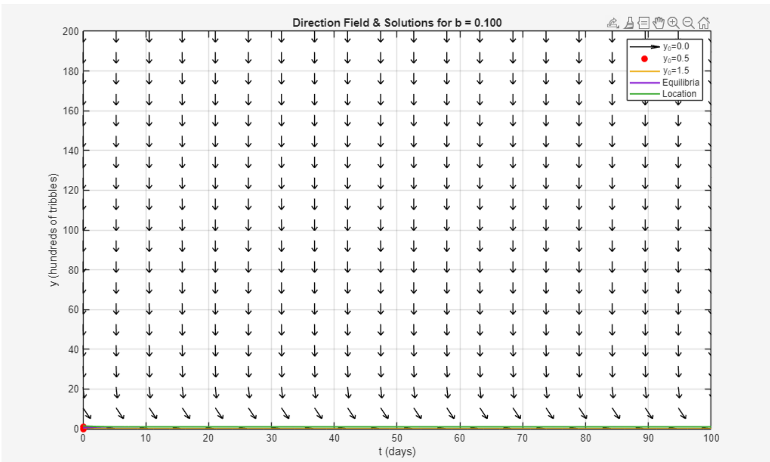

Figure 4: Direction field and solution trajectories for $b = 0.10$.

Question 8

Classification of equilibria. The equilibrium solutions y^* and their stability (from Figures 2–3.2) are:

b	y^*	Stability
0.005	0	unstable
	147.97	stable
0.05	0	unstable
	1.08	stable
	1.99	unstable
	12.63	stable
0.10	0	unstable
	0.97	stable

Basins of attraction. For each stable equilibrium y^*, the basin of attraction (the set of y_0 for which $y(t) \to y^*$ as $t \to \infty$) is:

$$
\begin{aligned}
b = 0.005: &\quad y^* \approx 147.97, \quad \text{basin } (0, \infty), \\
b = 0.05: &\quad \begin{cases} y^* \approx 1.08, & \text{basin } (0,\ 1.99), \\ y^* \approx 12.63, & \text{basin } (1.99,\ \infty), \end{cases} \\
b = 0.10: &\quad y^* \approx 0.97, \quad \text{basin } (0, \infty).
\end{aligned}
$$

Long-term behavior. Qualitatively, for each b:

- $b = 0.005$: Every positive initial population $y_0 > 0$ moves monotonically toward the single stable equilibrium $y^* \approx 147.97$.
- $b = 0.05$: There is bistability. If $0 < y_0 < 1.99$, the population tends to the low-level stable equilibrium $y^* \approx 1.08$; if $y_0 > 1.99$, it tends to the high-level equilibrium $y^* \approx 12.63$. Exactly at $y_0 = 1.99$ the system sits at an unstable equilibrium.
- $b = 0.10$: Regardless of $y_0 > 0$, all trajectories approach the unique stable equilibrium $y^* \approx 0.97$.

Question 9

Based on the behavior of the three tribble "species" with b-values from Question 6:

- White tribbles ($b = 0.05$) fail to recover from an initial population of 100 tribbles, showing a collapse (threshold effect).
- Gray tribbles ($b = 0.10$) always decline toward 100 tribbles no matter the initial condition.
- Brown tribbles ($b = 0.005$) reach a much higher equilibrium (around 14800 tribbles) and are stable if started from a safe population (e.g. 300 tribbles).

Based on analysis of the plots, Brown tribbles would be best for stocking the station. The direction field solution plot in Figure 2 showing the brown tribble conditions using b = 0.005, we can observe behavior of brown tribbles consistently being drawn towards the equilibrium of over 14,000 tribbles even from lower initial conditions. When compared to the white tribbles that are attracted to a lower equilibrium and gray tribbles that will always decline, the data clearly indicates superiority of brown tribbles. The equilibria table referenced in question 8 also support brown tribbles, indicating that when b = 0.005, all positive initial conditions go to the stable equilibrium of around 147.97 hundreds of tribbles, the highest of stable equilibria across the different tribbles. Using different starting conditions of tribble amounts ranging from few to many with various points in between show that the most ideal number of tribbles to start with is 300 tribbles, ensuring that the population reaches equilibrium and avoiding slow growth seen in smaller starting populations.

Question 10

(a) MATLAB code for seasonal model

See Appendix A for code

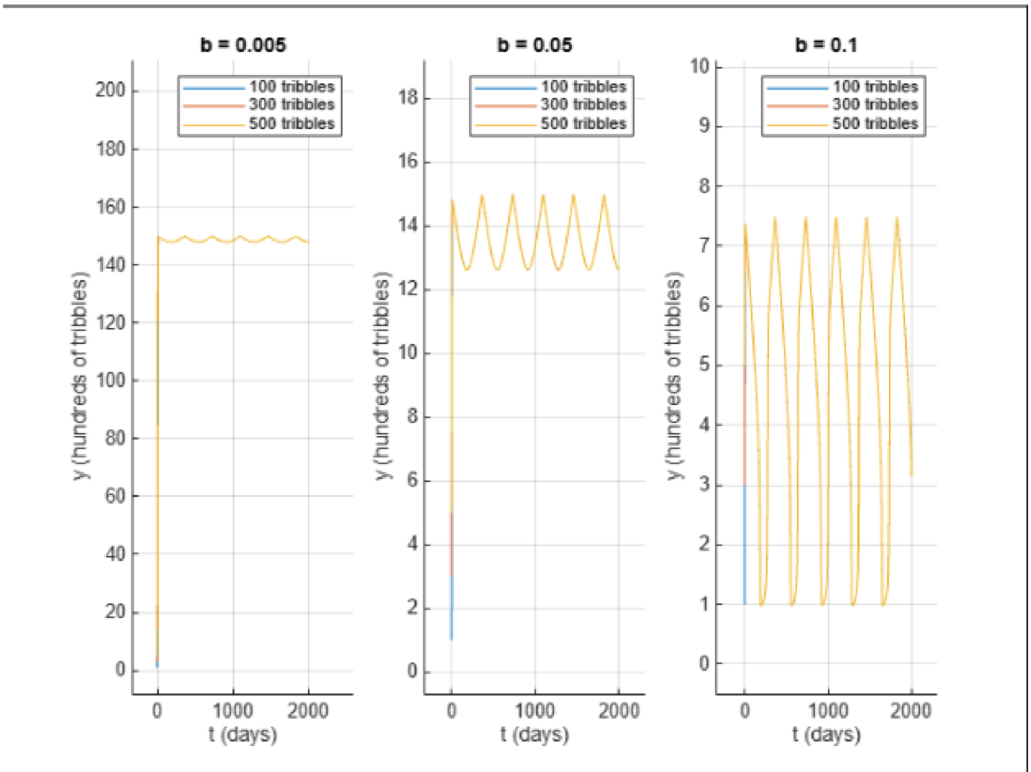

Figure 5: Seasonal model solution trajectories for b = 0.005, 0.05, and 0.10, showing the evolution of tribble populations (initially 100, 300, and 500) over 2000 days.

(b) Interpretation

- The system is now non-autonomous (explicit t-dependence), so solutions exhibit sustained oscillations (limit-cycle-like behavior) rather than converging to a constant equilibrium.
- All species show seasonal oscillations with period 365 days.
- No closed-form analytic solution exists; numerical integration (e.g. `ode45`) is required.
- Among the three:

  - White tribbles ($b = 0.05$) perform best, with moderate but stable oscillations.
  - Brown tribbles ($b = 0.005$) support large populations but remain sensitive to initial conditions.
  - Gray tribbles ($b = 0.10$) hover near low-population thresholds and risk extinction under adverse conditions.

Figure 6: Seasonal model solutions for $b = 0.005, 0.05, 0.10$ with initial populations of 100, 300, and 500 tribbles.

Question 11

Weaknesses of the model

- Assumes a spatially uniform population; does not account for clustering or sparse regions.
- No age structure—treats all tribbles as identical (no juvenile vs. adult dynamics).
- Single-species focus with a simplified harvesting function (fixed form, no variability).

Possible improvements

- Introduce stochastic terms for weather, hunting variability, or other uncertainties.
- Add spatial structure (e.g. a reaction-diffusion or agent-based grid model) to capture clustering and refuge zones.
- Incorporate age structure or stage-dependent reproduction/mortality rates.
- Allow time-varying or state-dependent carrying capacity to reflect environmental fluctuations.

While the techniques used to model the tribble populations are effective, they are not perfect and do not completely show all the interactions to be considered in an ecosystem. The system assumes a closed system of just the tribbles and the hunters, whereas in reality there would be many more interactions across many different species that would impact the tribbles and hunters differently. Additionally, the hunting function oversimplifies hunting by showing general trends that never change year to year and also doesn't differentiate between young and mature tribbles where in reality there would be differences in interaction and reaction based on maturity. To improve the model, accounting for this variation is paramount. Using more randomness to represent weather, variability in hunting, and variation of carrying capacity as a result can represent realistic entropy and the effects of those changes. Variation in reproduction rates can also more accurately represent tribble populations as opposed to assuming a constant average that does not vary with the season, population size, or resources.

3 Appendix

See Appendix A for Lengthy Work and Code

Code for 6. Graphical Analysis of Equilibrium Solutions

```
% Parameters
a = 0.75;
p = 1.5;
q = 1.25;
b_vals = [0.005, 0.05, 0.10];

% Range of y (in hundreds of tribbles)
y = linspace(0, 50, 1000);

% Function definition
f = @(y, b) a .* y - b .* y.^2 - (p .* y.^3) ./ (y.^3 + q);

% Plotting
figure;
hold on;
colors = lines(length(b_vals));

for i = 1:length(b_vals)
    b = b_vals(i);
    plot(y, f(y, b), 'DisplayName', sprintf('b = %.3f', b), ...
        'LineWidth', 1.5, 'Color', colors(i,:));
end

yline(0, '--k', 'LineWidth', 1);  % Horizontal axis (f(y) = 0)
xlabel('y (hundreds of tribbles)');
ylabel('f(y)');
title('Plot of f(y) = ay - by^2 - (py^3)/(y^3 + q) for various values of b');
legend('Location', 'northeast');
grid on;
xlim([0 50]);
ylim([-2 5]);
```

3.1 Code for 7

```
%Code for Question Seven
a = 0.75;
p = 1.5;
q = 1.25;
b_values = [0.005, 0.05, 0.10];

tspan = [0, 100];
ymin = 0; ymax = 200;
nt = 20; ny = 20;

for ib = 1:length(b_values)
    b = b_values(ib);

    f = @(t,y) a*y - b*y.^2 - p.*y.^3 ./ (y.^3 + q);

    T = linspace(tspan(1), tspan(2), nt);
    Y = linspace(ymin, ymax, ny);
    [TT, YY] = meshgrid(T, Y);

    S = f(TT, YY);
    U = ones(size(S));
    V = S;
    mag = sqrt(U.^2 + V.^2);

    figure;
    quiver(TT, YY, U./mag, V./mag, 0.5, 'k');
    hold on;

    y_scan = linspace(ymin, ymax, 1000);
    sig = sign(a*y_scan - b*y_scan.^2 - p*y_scan.^3./(y_scan.^3+q));
    idx = find(diff(sig)~=0);
    eqs = arrayfun(@(i) fzero(@(y) a*y - b*y.^2 - p*y.^3./(y.^3+q), y_scan(i)), idx);
    eqs = unique(eqs);

    plot(zeros(size(eqs)), eqs, 'ro', 'MarkerFaceColor','r');

    y0s = [min(eqs)*0.5, mean(eqs), max(eqs)*1.5];

    for y0 = y0s
        [t,y] = ode45(f, tspan, y0);
        plot(t, y, 'LineWidth',1.5);
    end

    title(sprintf('Direction Field & Solutions for b = %.3f', b));
    xlabel('t (days)');
    ylabel('y (hundreds of tribbles)');
    axis([tspan, ymin, ymax]);
    legend([arrayfun(@(y) sprintf('y_0=%.1f',y), y0s,'UniformOutput',false), ...
            'Equilibria','Location','best']);
    grid on;
    hold off;
end
```

3.2 Code for 10

```
%new project 1 question 10
clear
% Parameters
a = 0.75;
p = 1.5;
q = 1.25;
b_vals = [0.005, 0.05, 0.10];
% over 2000 days
tspan = [0 2000];
% Initial conditions
initial_conditions = [1, 3, 5];
% ODE options
options = odeset('AbsTol',1e-10,'RelTol',1e-7);
% Seasonal function
function dydt = seasonal_rhs(t, y, a, b, p, q)
   seasonal_factor = abs(sin(pi * t / 365));
   hunting = (p * y^3) / (q + y^3) * seasonal_factor;
   dydt = a*y - b*y^2 - hunting;
end
figure;
for k = 1:length(b_vals)
   b = b_vals(k);

   subplot(1,3,k);
   hold on;
   title(['b = ', num2str(b)]);

   for y0 = initial_conditions
       [t, y] = ode45(@(t,y) seasonal_rhs(t, y, a, b, p, q), tspan, y0,
options);
       plot(t, y, 'DisplayName', [num2str(100*y0), ' tribbles']);
   end

   xlabel('t (days)');
   ylabel('y (hundreds of tribbles)');
   legend;
   grid on;
end
```